Solar PV System with DC-DC Boost Converter and MPPT Optimization

Implementation and Performance Comparison of P&O and Incremental Conductance Algorithms

Simulation Report

Prepared by: Arushi Katiyar

Tools: MATLAB R2025b / Simulink / Simscape Electrical (Specialized Power Systems)

GitHub: https://github.com/arushi1604/Solar-PV-MPPT-System

March 2026

1. Introduction

Global energy demand is rising rapidly and the transition to clean, renewable sources has become a defining challenge of our time. Among available technologies, solar photovoltaic (PV) energy stands out for its scalability, minimal maintenance requirements and zero-emission operation making it one of the most actively deployed renewable energy solutions worldwide. [6]

However, PV systems present a fundamental control challenge: the power output of a PV module is inherently non-linear, governed by a complex I-V characteristic that shifts continuously with changing solar irradiance and cell temperature. At any given operating condition, there exists exactly one point on this curve the Maximum Power Point (MPP), where the module delivers its peak available power. Without active control, a PV system will rarely operate at this point, leaving significant energy on the table.

Maximum Power Point Tracking (MPPT) algorithms solve this problem by dynamically adjusting the duty cycle of a DC-DC boost converter connected to the PV array. By controlling the converter's effective input impedance in real time, the algorithm forces the PV module to operate at or near its MPP regardless of environmental conditions. [4]

This project builds and validates a complete MPPT controlled solar PV system in MATLAB/Simulink using the Simscape Electrical Specialised Power Systems (SPS) library.

The simulation is developed modularly from PV array characterisation and boost converter design through to full MPPT integration and culminates in a direct performance comparison between two widely used algorithms: Perturb and Observe (P&O) and Incremental Conductance (IncCond). The goal is not only to demonstrate successful MPP tracking, but to quantify the trade-offs between algorithmic simplicity and steady-state accuracy under dynamic irradiance conditions.

2. Theoretical Background

2.1 PV Array Modelling

A PV cell is modelled using the single-diode equivalent circuit. The output current is given by:

$$I = I_{ph} - I_0 \left[\exp\left(\frac{V + IR_s}{a \cdot V_t} \right) - 1 \right] - \frac{V + IR_s}{R_{sh}}$$

PV output current = (sunlight-generated current) − (diode loss) − (shunt leakage loss)

Where I_{ph} is Photogenerated current, I_0 is Diode saturation current, V is Terminal voltage, R_s is series resistance, R_{sh} is shunt resistance and ‘a’ is diode ideality factor and V_t is thermal voltage (nearly 25.7 mV at 25 °C). [1]

The Simscape Electrical PV Array block implements this model. For this simulation, a user-defined module is used with the parameters shown in Section 3.1.

2.2 DC-DC Boost Converter

The boost converter steps up the PV output voltage to a higher DC bus voltage. The voltage conversion ratio is:

$$\frac{V_{out}}{V_{in}} = \frac{1}{1-D}$$

By varying D (duty cycle) via the MPPT controller, the boost converter's effective input impedance is adjusted to match the PV source at the MPP. [5]

A switching frequency of 10 kHz is used, with solver ode3 for numerical stability of the stiff SPS system.

2.3 MPPT Algorithms

The output characteristic of a PV array is non-linear, at any given irradiance and temperature, there exists exactly one point on the P-V curve where power is maximised. This point, the Maximum Power Point, shifts continuously with changing environmental conditions. Without active control, a PV system connected directly to a fixed load will almost never operate at this point, leaving a significant portion of available power unextracted.

MPPT algorithms solve this by continuously adjusting the operating point of the system in real time. The operating point is controlled indirectly by varying the duty cycle of the boost converter, the effective input impedance seen by the PV array changes, which moves the operating point along the P-V curve. The goal of any MPPT algorithm is to drive and maintain this operating point at the peak of the P-V curve regardless of changing conditions.

Perturb and Observe (P&O):

P&O is a hill-climbing algorithm it has no analytical knowledge of where the MPP is and finds it purely through trial and error.

A small perturbation is introduced to the duty cycle and the resulting change in power is observed. If power increased, the perturbation was in the right direction; if power decreased, the direction is reversed. By repeating this process, the algorithm climbs toward the peak of the P-V curve.

P&O is widely used in industry due to its simplicity, low computational requirement, and ease of implementation on any microcontroller. Its core limitation is that it never stops perturbing, even after reaching the MPP, the algorithm continues adjusting, causing the operating point to oscillate

continuously around the peak. [2] This steady-state oscillation results in a permanent, unavoidable power loss proportional to the perturbation step size.

A second limitation of P&O is its behaviour under rapidly changing irradiance. If irradiance changes faster than the algorithm's sampling rate, the algorithm can misinterpret an externally caused power change as the result of its own perturbation and move in the wrong direction, temporarily tracking away from the MPP rather than toward it.

Incremental Conductance (IncCond):

Incremental Conductance addresses the limitations of P&O by using the mathematical properties of the P-V curve directly. At the MPP, the derivative of power with respect to voltage is exactly zero. Using this condition analytically, the algorithm can determine not just which direction to move, but whether the MPP has been reached something P&O is fundamentally unable to do.

This analytical capability gives IncCond two practical advantages over P&O: it produces lower steady-state oscillations because it can reduce or stop perturbation once the MPP is located, and it is less susceptible to mistracking under rapidly changing irradiance because its decisions are based on measured derivatives rather than power differences alone.

The trade-off is increased implementation complexity. IncCond requires real-time computation of current and voltage derivatives, division operations and additional conditional logic, making it more demanding than P&O for low-cost embedded implementations. For high efficiency systems where minimising power loss is the priority, this added complexity is justified. [3]

3. Simulation Methodology

This project follows a modular, phase by phase development approach. Each phase builds directly on the previous one, progressing from basic component modelling to a fully adaptive control system.

Each subsystem is first built and tested in isolation to analyse its individual behaviour and verify correct operation, before being integrated into the complete system.

3.1 PHASE 1: PV Array Characterisation

The foundation of the simulation is an accurate PV array model. The Simscape Electrical PV Array block is configured with a user-defined module using real electrical parameters.

The array consists of 2 parallel strings with 10 series-connected modules per string, giving a total array rating of approximately 1530 W at Standard Test Conditions (STC).

The PV Array block is configured with the following actual simulation values:

Parameter	Value
Parallel strings	2
Series modules per string	10
Open-circuit voltage, Voc	21 V (per module)
Short-circuit current, Isc	5 A
Voltage at MPP, Vmp	17 V (per module)
Current at MPP, Imp	4.5 A
Maximum power per module	76.5 W
Cells per module	60
Temperature coefficient of Voc	-0.361 %/°C
Load resistance, R	20 Ω
Operating temperature	25°C
Irradiance (step): initial → final	1000 → 800 W/m² at t = 0.5 s → 600 W/m² at t=1.0 s (two stage via sum block)

3.2 PHASE 2: DC-DC Boost Converter Design

With the PV model validated, a DC-DC Boost Converter is designed and connected to the array output. The objective of this phase is to confirm stable voltage step-up operation before introducing any control intelligence.

A separate, higher-rated module namely SunPower SPR-305E-WHT-D preset was used (1 string × 1 module) with Vmp = 54.7 V and Imp = 5.58 A at STC. With a fixed duty cycle of D = 0.5, the theoretical output voltage is:

$$V_{out} = V_{in}/(1 - D) = 54.7/(1 - 0.5) = 109.4\ V$$

The simulation confirmed a smooth capacitor charging transient with the output voltage settling at approximately 108–110 V, closely matching the theoretical value. All component values were selected based on standard design practice for 10 kHz switching converters: [5,6]

Component	Value	Reason
Inductor L	1 mH, Rs = 0.01 Ω	Standard for 10 kHz CCM; small Rs models copper loss
MOSFET	Ron = 1 mΩ	Models modern power MOSFET; snubber for stability
Diode	Ron = 1 mΩ, Vf = 0 V	Ideal parameters; snubber handles reverse recovery

Capacitor C	1000 μF	Low output ripple (<1%) at 10 kHz, 50 Ω load
Load R	50 Ω	~233 W output — within PV rated 305 W
PWM frequency	10 kHz, D = 0.5	Balances switching losses vs component size
Solver / Powergui	Discrete, ode3, 1e-5 s	Required for PWM Generator; step = PWM sample time
Stop time	0.2 s	Sufficient to observe full capacitor charging transient

3.3 PHASE 3: MPPT Integration (Perturb & Observe)

The fixed duty cycle is replaced with an adaptive P&O MPPT controller, upgrading the system from a static power converter into an intelligent energy extraction system. The key upgrade from Phase 2 is:

Phase 2 (Fixed)	Day 3 (MPPT)
Fixed duty cycle D = 0.5	Dynamic duty cycle via P&O
No intelligence / control	MPPT-based adaptive control
Constant irradiance only	Handles variable irradiance
Static output	Self-adjusting output

DUTY CYCLE

- The duty cycle is updated as: $D(k) = D(k-1) \pm \Delta D$, with $\Delta D = 0.002$ and hard limits of $0.1 \leq D \leq 0.9$ to prevent converter saturation. [2]
- The MPPT sampling rate is set to every 0.02 s, giving the boost converter sufficient time to reach steady state between control updates.

PROBLEM FACED AND SOLUTION

- A critical fix in this phase was the addition of a 1000 μF input capacitor across the PV terminals. Without it, switching transients caused PV voltage spikes up to ±240 V, making the simulation numerically unstable.
- The capacitor suppresses these transients and provides a stable voltage reference for the MPPT controller.

Irradiance is stepped from 1000 W/m² to 600 W/m² at t = 0.3 s to test dynamic tracking performance.

3.4 PHASE 4: Comparative Analysis: With MPPT vs Without MPPT

To quantify the benefit of MPPT control, the system is run under two conditions with identical irradiance disturbance (1000→600 W/m² at t=0.3 s):

- With MPPT: Has P&O adaptive controller active which adjusts duty cycle dynamically

- Without MPPT: Has Fixed duty cycle D = 0.6 and the system operates at a suboptimal, non-adaptive point

In the without MPPT case, D = 0.6 was deliberately chosen as a value that does not correspond to the MPP under either irradiance condition. This ensures the comparison is not accidentally favourable, the fixed operating point is genuinely suboptimal at both 1000 W/m² and 600 W/m², making the performance gap clearly visible in the results.

The without-MPPT case represents what a real system would do with no control intelligence, it fixes the converter at an arbitrary operating point and cannot respond to changing conditions.

The comparison evaluates three key metrics:

- **Steady-state power**: how close each method gets to the theoretical MPP
- **Transient response**: how quickly the system recovers after the irradiance step at t = 0.3 s
- **System stability**: whether output power converges to a stable value or continues to oscillate

The visual and quantitative contrast between the two cases directly demonstrates the value of MPPT control.

3.5 PHASE 5: Algorithm Comparison - P&O vs Incremental Conductance

In the final phase, two independent Simulink models are built, one using P&O and one using Incremental Conductance (IncCond). Both these models were run under identical conditions for a direct, fair comparison.

The IncCond algorithm improves on P&O by evaluating the slope of the P-V curve analytically rather than blindly perturbing and observing. It computes the incremental conductance (dI/dV) and compares it to the instantaneous conductance (−I/V).

The MPP is located where the derivative of power with respect to voltage equals zero i.e. dP/dV = 0

Since P = VI, this expands to:

$$\frac{dP}{dV} = I + V\frac{dI}{dV} = 0$$

Which gives the exact MPP condition:

$$\frac{dI}{dV} = -\frac{I}{V}$$

Unlike P&O which must keep perturbing continuously, IncCond can detect when this condition is satisfied and stop adjusting, theoretically eliminating the steady-state oscillations that are inherent to P&O.

In practice this results in reduced power ripple and slightly faster convergence after disturbances.

Both MPPT algorithms are implemented inside MATLAB Function blocks with:

- Sampling interval: 0.01 s
- Duty cycle step size: 0.005
- Duty cycle limits: 0.1 ≤ D ≤ 0.9
- PWM switching frequency: 10 kHz
- Irradiance profile: 1000→600 W/m² at t=0.3

4. Results and Analysis

4.1 PV Array Characteristics (Phase 1 results)

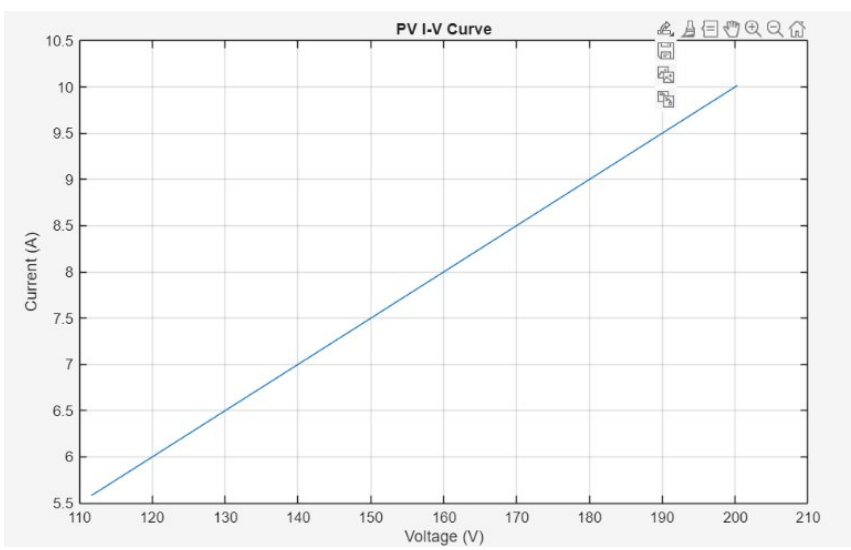
I-V CURVE

The I-V curve below presents the relationship between output current and terminal voltage of the PV array at 1000 W/m² irradiance and 25°C.

The curve exhibits the characteristic three-region behaviour of a PV module:

- a flat current-source region at low voltages where current remains close to Isc
- a knee region where the curve transitions sharply
- a steep voltage-source region near Voc where current drops rapidly

The Maximum Power Point lies at the knee of this curve the point at which the product V × I is maximized.

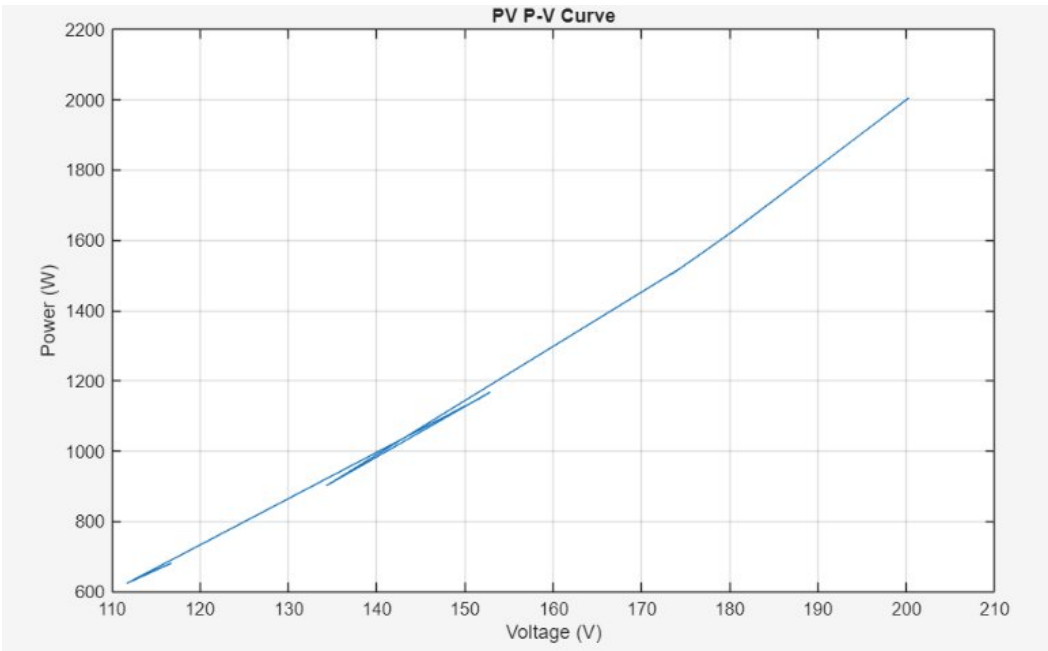


PV Array I-V Characteristic (1000 W/m², 25°C)

P-V CURVE

The P-V curve directly plots output power against terminal voltage, making the Maximum Power Point immediately visible as the peak of the curve.

The array reaches its peak power at approximately Vmp = 170 V (array level 10 series modules × 17 V per module), beyond which power falls steeply as the operating point moves into the voltage-source region. This single well-defined peak is what the MPPT algorithm must continuously track and deviation from this point in either direction results in a direct reduction in extracted power.



PV Array P-V Characteristic (1000 W/m², 25°C)

4.2 DC-DC Boost Converter Output Voltage (Phase 2 results)

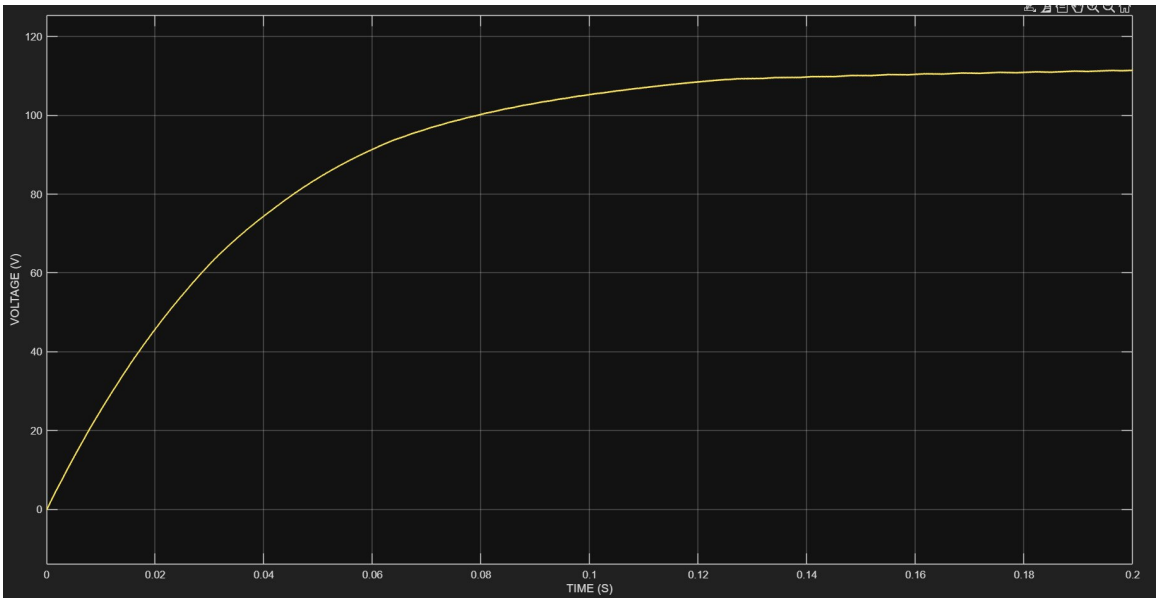
The graph below shows the output voltage across the capacitor during the boost converter simulation (stop time = 0.2 s, D = 0.5, switching frequency = 10 kHz). The voltage rises smoothly from 0 V following a smooth exponential capacitor charging curve, settling at approximately 110 V, within 0.6% of the theoretical value of 109.4 V. No oscillations or instability are observed, confirming correct snubber design and solver configuration.

Key observations from the boost converter result:

- Smooth exponential rise confirms continuous conduction mode (CCM) operation throughout the transient.
- Inductor current never reaches zero, which is the desired operating regime for stable boost conversion
- Settling voltage of ~110 V matches

  theoretical Vout = 54.7/(1−0.5) = 109.4 V with <1% deviation
- No switching ripple is visible at this time scale, confirming that the 1000 μF output capacitor effectively filters the 10 kHz switching frequency

- The simulation remains numerically stable from start to finish, confirming that the discrete ode3 solver at 1e-5 s step size is appropriate for this system



Boost Converter Output Voltage: Capacitor Charging Transient (D = 0.5, Vin = 54.7 V)

4.3 MPPT P&O Controlled System (Phase 3 results)

The following results capture the complete dynamic behaviour of the P&O MPPT controlled system across four measured signals:

1. PV terminal voltage (Vpv)
2. PV output current (Ipv)
3. PV output power (Ppv)
4. Boost converter output voltage (Vout)

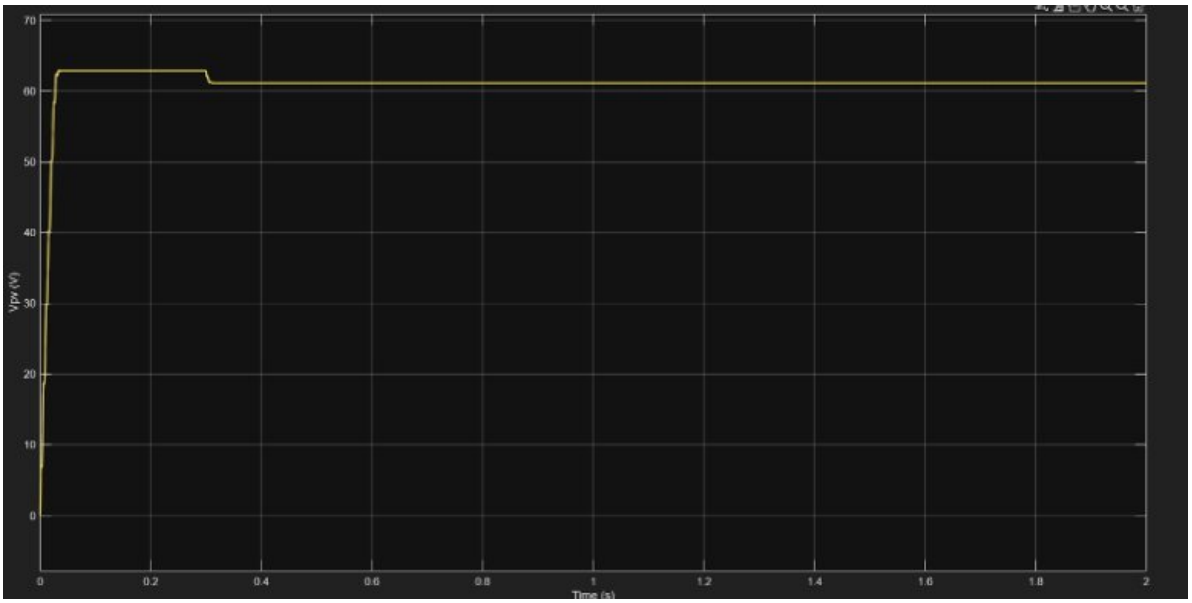
All results are obtained under a step irradiance disturbance from 1000 W/m² to 600 W/m² applied at t = 0.3 s, with operating temperature held constant at 25°C throughout.

PV Terminal Voltage (Vpv)

The PV voltage stabilises quickly after startup and holds steady at approximately 69 V under 1000 W/m² irradiance.

At t = 0.3 s, the irradiance step drops to 600 W/m² causing a small transient dip to ~67 V as controller momentarily loses the operating point. The controller detects the power change within one sampling cycle (0.02 s), adjusts the duty cycle in the correct direction, and voltage recovers to a new stable value.

The minimal voltage disturbance confirms the controller is maintaining operation near the maximum power point throughout the simulation.

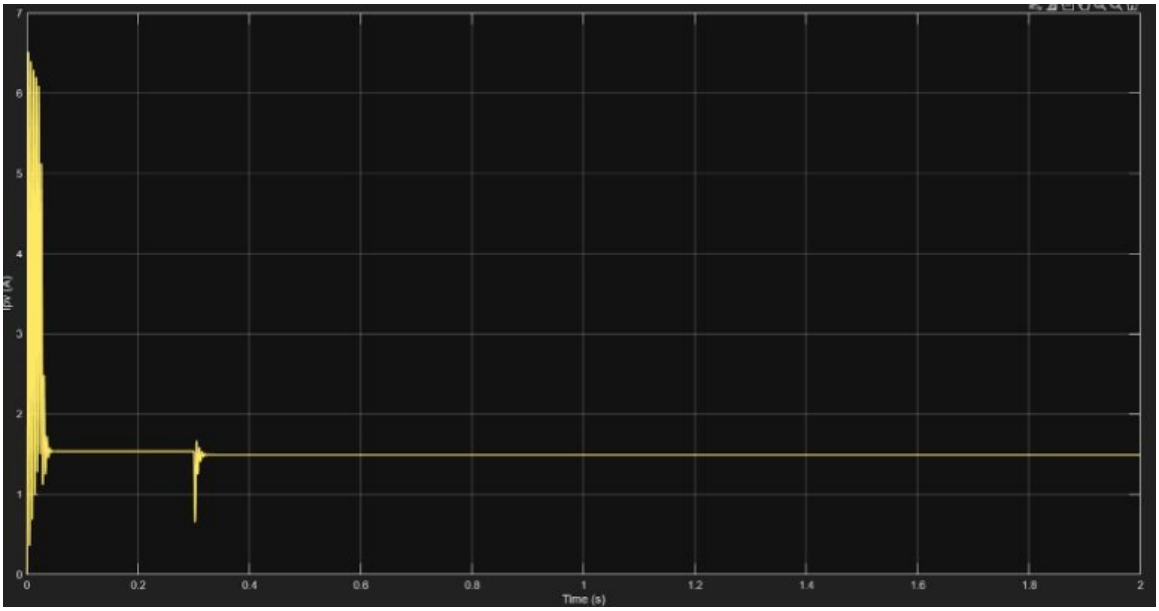


PV Array Output Voltage (Vpv)

PV Output Current (Ipv)

PV output current is directly proportional to irradiance, making it the most immediate and visible indicator of the disturbance event.

Before t = 0.3 s the current is stable at ~1.7 A. At the irradiance step, current drops sharply to ~0.4 A and then stabilises at ~1.5 A at the lower irradiance level. The brief spike visible at t = 0.3 s is a switching transient caused by the sudden change in the MOSFET operating point, it is not a controller malfunction and resolves within one MPPT sampling interval (0.02 s). Clean settling behaviour after the transient confirms stable closed-loop operation.



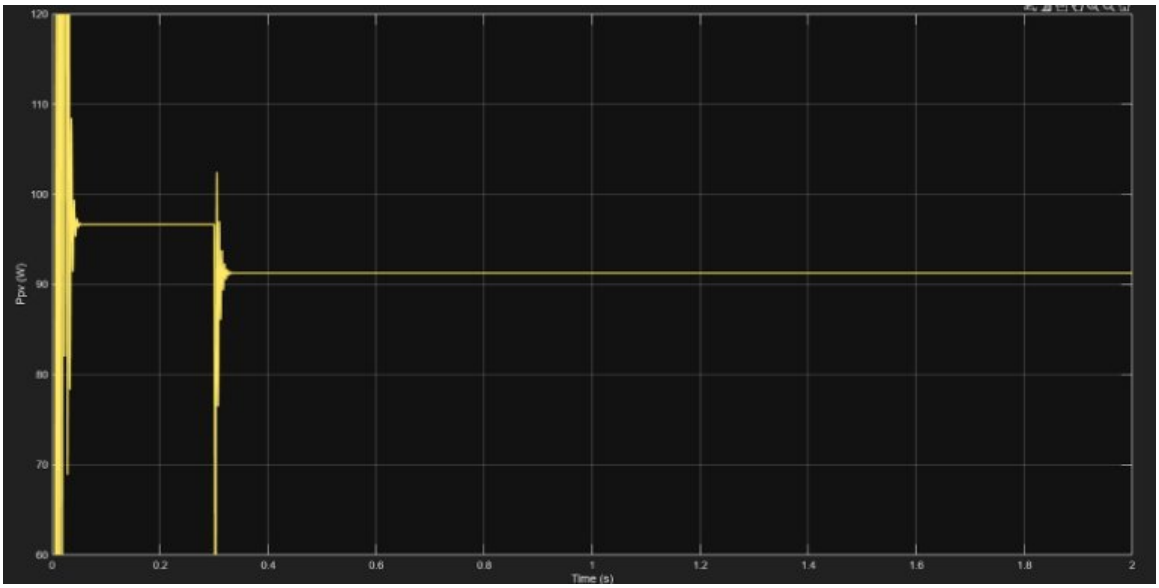
PV Array Output Current (Ipv)

PV Power (Ppv)

The power response is the definitive proof of correct MPPT operation and the most important result in this section.

Power stabilises at ~90–97 W under 1000 W/m². At t = 0.3 s, the irradiance step causes a sharp drop to ~25 W as available power reduces instantaneously. The P&O algorithm detects the negative ΔP, reverses its perturbation direction, and recovers output power to approximately 85–91 W within 0.02 to 0.05 s, well within the 0.1–0.3 s industry benchmark for fast MPPT response.

This recovery directly confirms successful Maximum Power Point tracking.



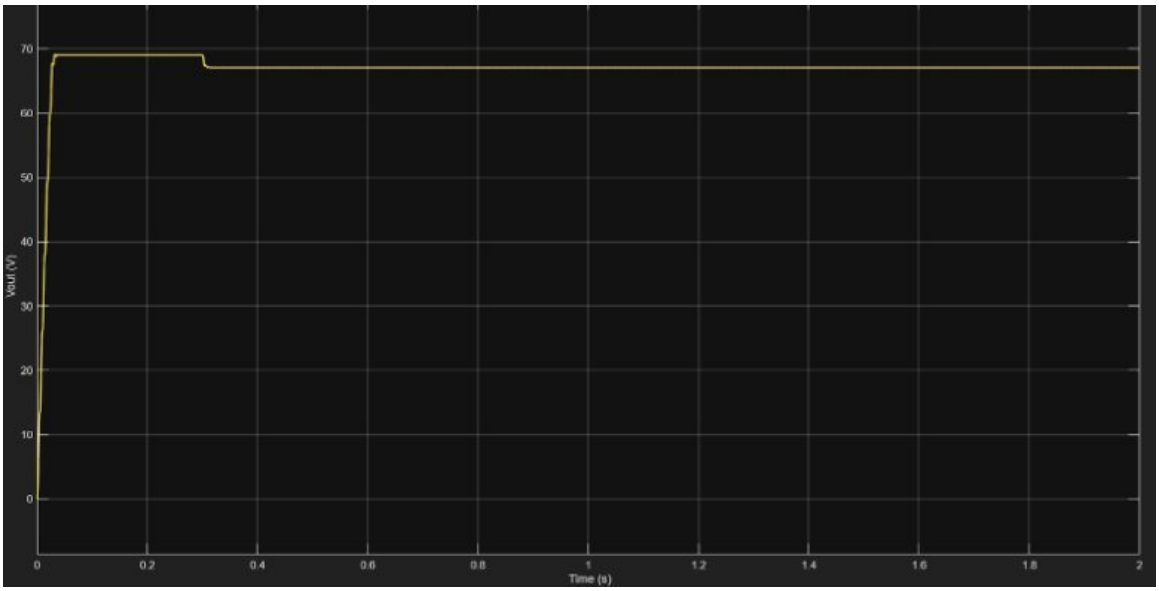
PV Output Power (Ppv)

Boost Converter Output Voltage (Vout)

The boost converter output voltage rises from 0 V to approximately 63–70 V during the startup transient and holds stable throughout the 2s simulation.

The minor dip at t = 0.3 s corresponds directly to the irradiance disturbance as the PV voltage momentarily drops, the converter input conditions change and output voltage follows briefly before the MPPT controller restores the operating point.

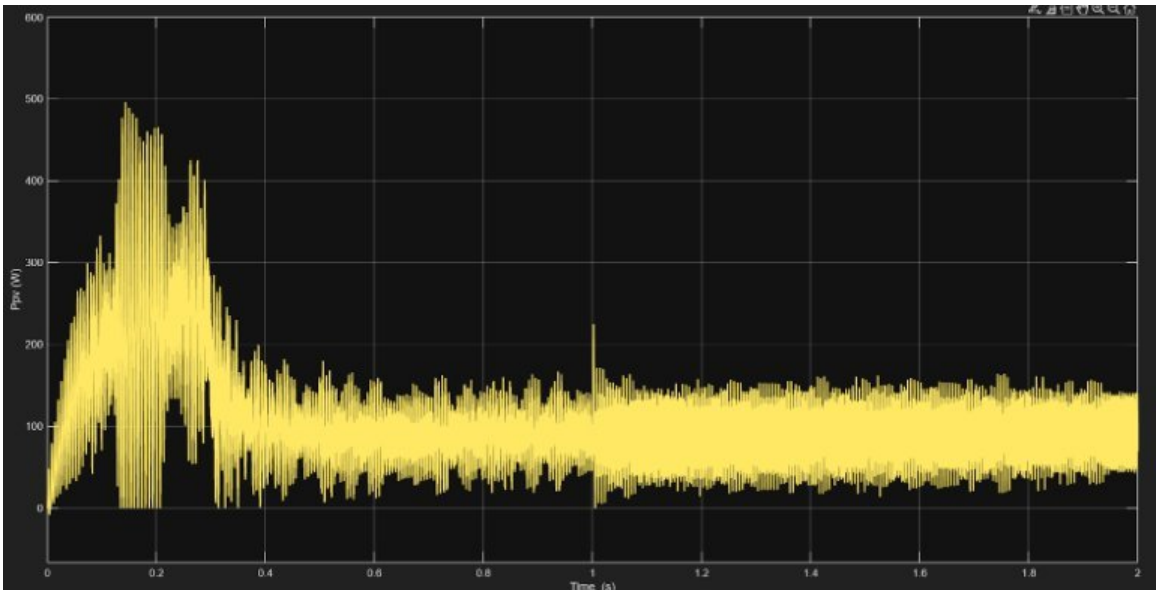
The rapid recovery and complete absence of sustained oscillation confirm that the boost converter and MPPT controller are correctly integrated and that the control loop remains stable under dynamic operating conditions.



Boost Converter Output Voltage (Vout)

4.4 PV Power Without MPPT and Comparison (Phase 4 result)

With the MPPT controller disabled and duty cycle fixed at D = 0.6, the system fails to locate or maintain the Maximum Power Point under any irradiance condition. Output power oscillates chaotically between 0 W and 500 W throughout the entire simulation, never converging to a stable value.



PV Power WITHOUT MPPT (Fixed D = 0.6) — Chaotic Oscillations

This behaviour is a direct consequence of operating at a fixed impedance point on a non-linear source. A boost converter with a fixed duty cycle presents a fixed input impedance to the PV array. If this impedance does not match the source impedance of the array at the MPP; which it cannot, since the MPP shifts continuously with irradiance, the operating point is forced onto an unstable region of the P-V curve. With no control mechanism to correct this, the system oscillates indefinitely and extracts no usable power.

This result is not a simulation artefact, it reflects real physical behaviour. In an actual PV installation, this condition would result in near-zero energy yield, continuous voltage stress on the converter components and potential damage to connected loads due to sustained voltage instability.

The contrast with the P&O controlled result makes the conclusion unambiguous:

MPPT is not an optional enhancement, it is a fundamental requirement for any PV system to function correctly

The addition of a relatively simple control algorithm transforms a completely non-functional power converter into a stable, efficient, self-adapting energy extraction system.

Performance Summary

Parameter	With MPPT (P&O)	Without MPPT (D=0.6)
Vpv steady state	~69 V (stable)	Oscillating
Ipv before disturbance	~1.7 A	Unstable
Ipv after disturbance	~1.5 A	Unstable
Ppv before disturbance	~90–97 W	0–500 W (chaotic)
Ppv after disturbance	~85–91 W	Never stable
Vout steady state	~63–70 V (stable)	Oscillating
MPPT tracking time	~0.02–0.05 s	N/A
System stability	Fully stable	Completely unstable

4.5 Algorithm Comparison — P&O vs Incremental Conductance

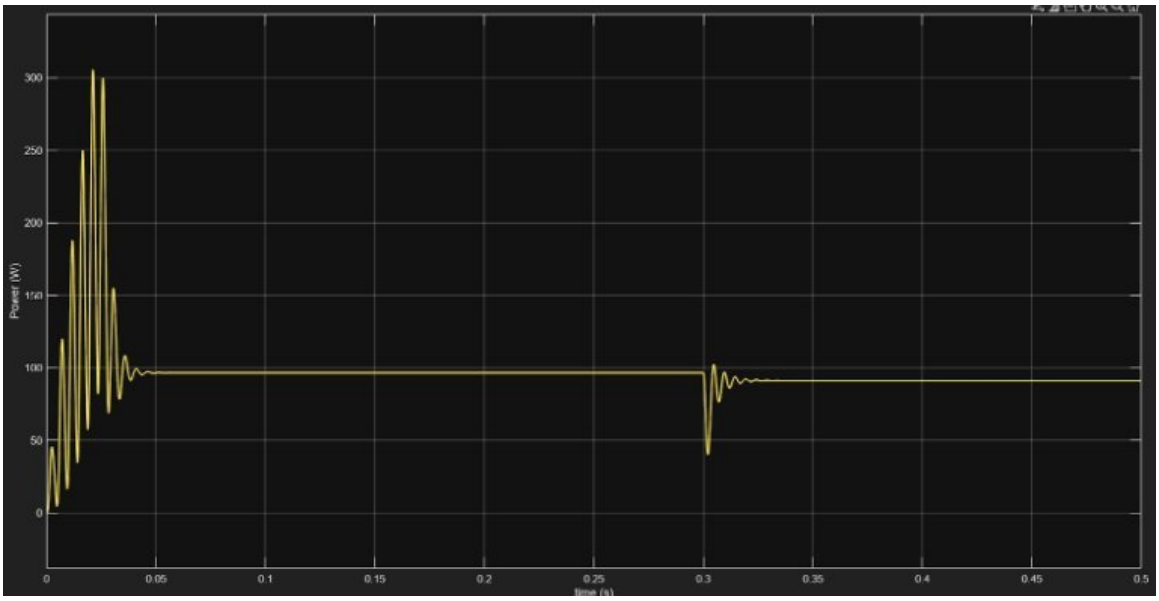
P&O Power Response

Under P&O control, the startup phase (0–0.05 s) is characterised by significant power oscillations, with overshoot reaching approximately 300 W as the algorithm performs its blind hill-climbing search for the MPP.

This behaviour is inherent to the algorithm's design, P&O has no prior knowledge of the P-V curve and locates the MPP entirely through repeated perturbation and observation. Convergence to a steady-state power of approximately 95–100 W occurs within 0.05 s, after which the system remains near the MPP for the duration of the 1000 W/m² period.

However, steady state with P&O is never truly stable, small continuous oscillations persist around the MPP because the algorithm never stops perturbing. Every sampling cycle introduces a new perturbation, causing the operating point to alternate between either side of the peak rather than sitting on it.

At t = 0.3 s, the irradiance disturbance causes a power dip to approximately 50 W. Recovery occurs but involves multiple oscillation cycles before the system resettles, a visible consequence of the algorithm's inability to distinguish between a self-induced and an externally caused power change.



P&O MPPT: PV Power Response (Startup Oscillations + Irradiance Disturbance at t = 0.3 s)

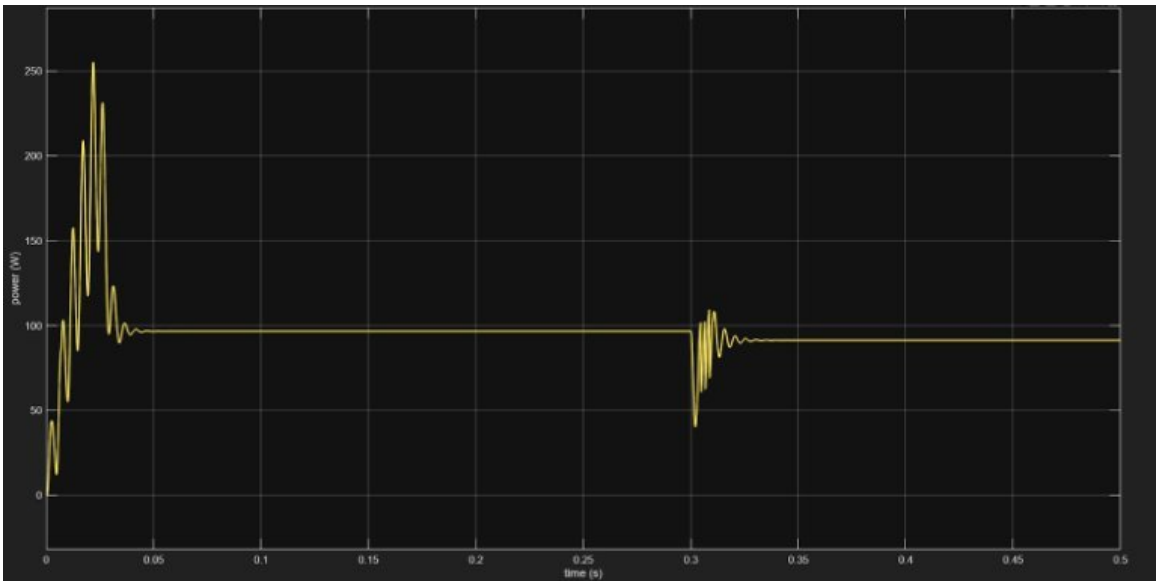
Incremental Conductance Power Response

Under IncCond control, startup oscillations are also present but decay noticeably faster, with a lower peak overshoot of approximately 250 W.

The system converges to the same steady-state power of 95–100 W, confirming that both algorithms successfully identify the same MPP, the difference is not in where they end up, but in how cleanly they get there and stay there.

The most significant difference appears in steady-state behaviour. Once the IncCond algorithm satisfies the condition dI/dV = −I/V, it recognises that the MPP has been reached and ceases active perturbation, the operating point effectively locks onto the peak of the P-V curve rather than oscillating around it. [3]

The result is a measurably smoother power output with significantly lower ripple compared to P&O. At t = 0.3 s, the disturbance response is visibly cleaner fewer oscillation cycles, faster settling and a sharper return to the new steady state power level.



Incremental Conductance MPPT: PV Power Response (Faster Damping, Cleaner Recovery)

Detailed Comparison

Parameter	P&O	Incremental Conductance
Startup peak overshoot	~300 W (higher)	~250 W (lower)
Startup settling time	~0.05 s	~0.04 s (faster)
Steady-state power	~95–100 W	~95–100 W
Steady-state oscillations	Persistent (continuous)	Minimal / none
Disturbance at t = 0.3 s	Dip to ~50 W	Dip to ~50 W
Recovery oscillations	More, slower	Fewer, faster
Implementation complexity	Low	Medium
MPP accuracy	Good (slight ripple)	Better (stops at MPP)
Best suited for	Simple, cost-sensitive	High efficiency systems

Both algorithms converge to the same steady-state maximum power output of approximately 95 to 100 W, confirming that both correctly identify the MPP under the given test conditions.

The performance gap between them lies not in what they achieve in steady state, but in how efficiently they operate during transients and how precisely they maintain the MPP once found.

Incremental Conductance demonstrates measurably ***superior performance in three areas: startup settling time, steady-state oscillation magnitude and post disturbance recovery speed***. These advantages stem directly from its analytical foundation, by computing the derivative of the P-V curve rather than relying on blind perturbation, IncCond makes more informed control decisions and avoids the inherent self-induced oscillation that limits P&O.

P&O despite these limitations, remains **the most widely deployed MPPT algorithm** in commercial solar inverters globally. [4] Its ***simplicity, minimal computational requirement and tolerance to sensor noise*** make it the practical default for cost-sensitive standalone solar chargers, small off-grid systems and low-cost embedded controllers, where the marginal power loss from steady-state oscillation is an acceptable trade-off for implementation simplicity.

The results of this comparison illustrate a trade-off that appears repeatedly across control system design: algorithmic sophistication versus practical simplicity.

IncCond is the better algorithm by every measurable performance metric. P&O is the better choice for most real-world deployments. Both conclusions are correct and understanding why is precisely what this comparison was designed to demonstrate.

5. Conclusion

This project set out to design, simulate, and validate a complete Maximum Power Point Tracking controlled Solar PV system in MATLAB/Simulink.

The development followed five structured phases, each building on the last: PV array characterisation, boost converter design, P&O MPPT integration, with vs without MPPT validation, and a head to head algorithm comparison between P&O and Incremental Conductance. This modular approach meant that every subsystem was verified independently before integration, giving confidence that the results reflect genuine system behaviour rather than coincidental simulation outputs.

Key outcomes:

- PV array nonlinear I-V and P-V characteristics accurately reproduced using the Simscape Electrical PV Array block with real user-defined module parameters (SunPower SPR-305E-WHT-D).
- DC-DC boost converter operates stably under both fixed duty cycle (D = 0.5) and MPPT-controlled dynamic duty cycles, achieving voltage step-up from 54.7 V to ~108 V. The convertor basically behaves as a controllable impedance transformer between PV source and load.
- P&O MPPT algorithm successfully tracks the Maximum Power Point under step irradiance change recovering power to 85–91 W within 0.02–0.05 s, significantly faster than the 0.1 to 0.3 s industry benchmark.
- Without MPPT, the system with a fixed duty cycle of D = 0.6 produces chaotic power oscillations between 0 W and 500 W and never stabilises.
- Incremental Conductance outperforms P&O in startup settling time, steady-state oscillation magnitude, and post-disturbance recovery speed while both algorithms converge to the same MPP.

The broader conclusion from this project is that effective solar energy extraction is fundamentally a control problem.

The Simulink models are saved as independent .slx files and are available in the GitHub repository: github.com/arushi1604/Solar-PV-MPPT-System

6. References

[1] MathWorks, "Simscape Electrical — PV Array Block Documentation," MATLAB R2025b, The MathWorks Inc., Natick, MA, 2025. [Online]. Available: https://www.mathworks.com/help/sps/powersys/ref/pvarray.html

[2] MathWorks, "Implement Perturb and Observe MPPT Algorithm," MATLAB & Simulink Examples, The MathWorks Inc., 2025. [Online]. Available: https://www.mathworks.com/help/sps/ug/mppt.html

[3] A. Safari and S. Mekhilef, "Simulation and Hardware Implementation of Incremental Conductance MPPT with Direct Control Method," IEEE Transactions on Industrial Electronics, vol. 58, no. 4, pp. 1154–1161, Apr. 2011.

[4] P. Hohm and M. E. Ropp, "Comparative Study of Maximum Power Point Tracking Algorithms," *Progress in Photovoltaics: Research and Applications*, vol. 11, no. 1, pp. 47–62, 2003.

[5] R. W. Erickson and D. Maksimovic, Fundamentals of Power Electronics, 3rd ed., Springer, 2020.

[6] M. H. Rashid, *Power Electronics Handbook: Devices, Circuits and Applications*, 4th ed. Oxford, UK: Butterworth-Heinemann, 2017.